

Where Should a Mobile Manipulator Look?

A Reproducible Benchmark for Deviation-Seeking As-Built Verification

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Abstract—As-built verification asks whether a building matches its model. Existing pipelines collect data along preset or coverage-driven routes and compare offline, so the verification objective never informs where the robot looks. We ask whether making deviation detection the planning objective actually pays, and we build the apparatus needed to answer that: an open benchmark that derives Gazebo worlds from building models, injects deviations of known type, location and magnitude at a controlled density, and scores a run by detections per metre of drivable traversal. On a Clearpath Husky A300 with a UR5e and a wrist-mounted RGB-D camera, we compare a greedy information-gain planner that selects both base poses and arm configurations against coverage, frontier and goal-directed baselines. Three findings stand out. Measuring distance around walls rather than in straight lines reverses the ranking, so the accounting is not a detail. Under that honest accounting, greedy information-gain planning does not beat systematic coverage. Including the manipulator in viewpoint selection helps every planner that stops to look, by up to 9.3 percentage points. We release the benchmark so that the second finding can be attacked directly.

I. INTRODUCTION

Building Information Models (BIM) are increasingly used as a geometric prior for construction robots [1]. A natural application is as-built verification: driving a sensor through a partially finished building and reporting where reality departs from the model. Established pipelines capture along a preset or coverage-driven route and register against the model afterwards [2], [3], [4]. The verification objective therefore never influences where the robot goes.

Making that objective drive the plan is an appealing idea, and it is the one we set out to test. A robot that knows which elements it has already resolved should be able to spend a limited traversal budget where uncertainty is highest, rather than sweeping a floor uniformly. Including a manipulator adds a second lever: an arm reaches vantage points a fixed sensor cannot, looking beneath assemblies and into equipment enclosures without repositioning the base.

Testing the idea required more infrastructure than the idea itself: no shared benchmark supplies exact deviation ground truth at a controllable density, an observation model fast enough for a planner loop, and a traversal metric that accounts for walls. This paper contributes that apparatus and reports what it says.

Contributions. (i) An open, reproducible benchmark: element manifests keyed by IFC GlobalId, seeded de-

viation sampling at a chosen density, matched Gazebo worlds and exact labels, and a geometric observation model costing ~ 12 ms per candidate viewpoint. (ii) A viewpoint-planning formulation over base poses and arm configurations, with camera pose derived from the generated robot description so the sensor mount cannot silently diverge from the robot. (iii) An evaluation showing that traversal accounting reverses planner ranking, that greedy information gain does not beat coverage under drivable distance, and that the manipulator helps every planner tested.

II. RELATED WORK

As-built verification. Bosché [2] registers laser scans against CAD objects for dimensional compliance; Turkan et al. [3] and Golparvar-Fard et al. [4] track progress by comparing captured data with 4D models. These are offline comparisons of data gathered along routes chosen for coverage, not for the questions the comparison will ask.

Active perception. Next-best-view planning dates to Connolly [5] and remains active for volumetric reconstruction and exploration [6], [7]. Frontier exploration [8] maximises newly observed free space. These objectives reward seeing more, whereas verification rewards resolving specific modelled elements. Recent construction work does close the perception–action loop, but treats model–reality divergence as a safety signal while reaching a navigation goal rather than as the objective itself [17].

Budgeted selection. Choosing viewpoints under a travel budget is an orienteering problem [9]; greedy benefit-per-cost is the standard heuristic and carries guarantees when the objective is submodular [10], [11]. Our results suggest the greedy routing, not the objective, is what limits performance here.

III. BENCHMARK

A. Scenes and deviations

An adapter converts IFC products into a manifest of elements, each retaining its GlobalId, IFC class, pose and geometry (Fig. 1). Because that adapter currently approximates each product by its world-aligned bounding box, a wall with a door becomes a solid barrier, which inflates occlusion exactly where a viewpoint planner’s reasoning lives. For the experiments here we therefore generate a synthetic storey from primitives: a central corridor with rooms down both sides, with walls emitted as jambs and a lintel so that a doorway is a real opening.

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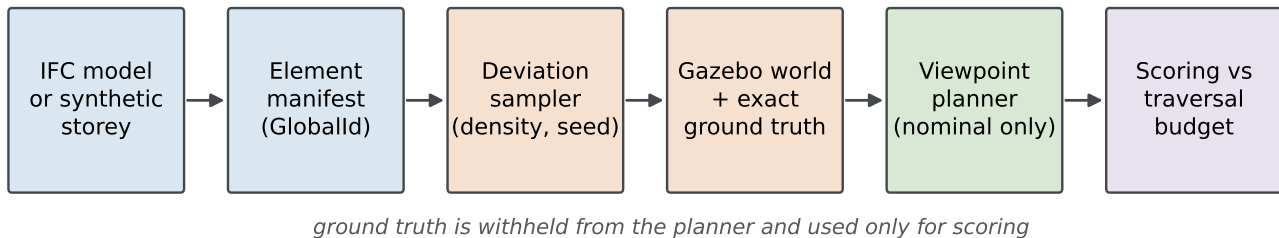


Fig. 1. The benchmark pipeline. A building model becomes an element manifest keyed by IFC GlobalId; a seeded sampler injects deviations at a chosen density; the same generator emits the Gazebo world and the exact labels, so scene and ground truth cannot drift apart. Planners see only the nominal model.

The default layout has 110 elements. Contents sit at heights that reward different arm postures: services near the ceiling, panels at chest height, and fittings tucked against the far side of equipment.

A seeded sampler injects five deviation types — displaced, rotated, resized, missing and unexpected — at a specified density, defined as the fraction of nominal elements carrying a deviation. Unexpected elements are counted separately, since they add to the scene rather than modifying an element. A scenario identifier and seed reproduce a benchmark exactly.

B. Observation model

Rendering every candidate viewpoint is infeasible when a planner scores thousands of them, so we answer the cheaper question a planner actually asks: what fraction of an element’s surface would this camera see? Elements are treated as oriented boxes and their surfaces sampled; a sample counts when it lies inside the view frustum, faces the camera within an incidence limit, and is unoccluded by another element. Evaluating one viewpoint against 110 elements costs about 12 ms.

Each deviation type is revealed by a different observation. A missing element is found by seeing into its modelled volume and finding it empty; rotated, resized and unexpected elements by observing the as-built element; a displaced element by either. A deviation counts as detected the first time its governing observation exceeds a visible-fraction threshold, which stands in for detector sensitivity.

This model assumes perfect recognition and localisation, so it measures viewpoint quality rather than perception robustness: the right separation for studying where to look, and a limitation for hardware transfer.

C. Traversal metric

Detections per unit distance is the quantity we want to improve, so the denominator must be a distance the robot could actually drive. We compute it on an inflated occupancy grid using the platform’s own Nav2 settings [12]: a 0.06 m costmap resolution and the 0.495×0.349 m footprint, inflated by its 0.606 m circumscribed radius. The platform’s global planner is NavFn, which is Dijkstra

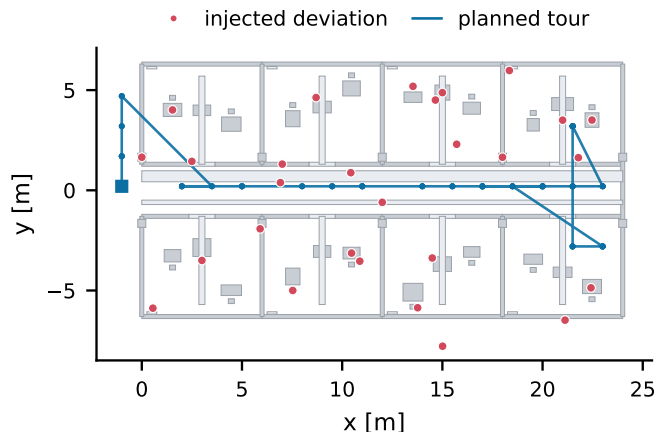


Fig. 2. Plan view of the synthetic storey with one sampled scenario. Grey rectangles are elements, light grey are overhead services, red markers are injected deviations, and the blue path is the tour selected by the deviation-seeking planner under a 60 m drivable budget. The planner concentrates on the corridor and enters comparatively few rooms.

over a costmap, so we compute that quantity directly rather than through an action server; a round trip per query would make evaluation intractable. Treating inflated cells as lethal bounds the planner’s answer from above.

The choice is not cosmetic. On this storey, drivable distance averages $1.8\times$ the straight-line distance and reaches $11\times$.

IV. PLANNING AND BASELINES

Candidate viewpoints pair a base pose, drawn from a grid of navigable free space, with one of five inspection arm postures. Camera pose comes from forward kinematics over the generated robot description, so the arm mount and camera transform cannot diverge from the robot; we verified the chain against the running simulator’s TF tree to machine precision. The fixed-sensor configuration is the same generator with the arm frozen, so the ablation compares like with like.

Every planner sees only the nominal model. Ground truth is used afterwards, by the scorer, which is what prevents a planner from quietly cheating.

The deviation-seeking planner maintains an independent Bernoulli belief over each element’s deviation state. A viewpoint is worth the expected entropy it removes, where an observation’s resolving power ramps from zero below the detection threshold to one when the element is properly seen. Viewpoints are chosen greedily by benefit per unit distance under a traversal budget.

Baselines are coverage (visit every base pose in a nearest-neighbour sweep), frontier (greedily reveal unseen area [8]), and goal-directed (drive to a distant goal, observing only from poses along the way). All four share the same candidate set, budget and cost function.

V. EXPERIMENTS

We evaluate in Gazebo on a Clearpath Husky A300 with a UR5e and a wrist-mounted RealSense D435, configured through the vendor’s own robot.yaml generator so the same description serves simulation and hardware. Scenes use the 110-element storey with deviations sampled at density 0.2 plus a 0.05 unexpected rate, giving 28 deviations for the seed shown. Each planner receives a 60m budget and a 5m usable depth range, and we report the mean over five deviation seeds.

A. Traversal accounting changes the answer

Table I and Fig. 3 report the same runs under both cost models. Under straight-line distance the deviation-seeking planner leads coverage by 2.1 points, which is inside one standard deviation. Under drivable distance it trails by 6.4 points. The goal-directed baseline loses 17 points when walls are charged for, because a straight run between distant poses is precisely what a building does not allow.

We consider this the most transferable result in the paper: a detections-per-metre claim is only as good as the metre, and straight-line distance systematically flatters planners that wander.

B. Greedy information gain does not beat coverage

Under honest accounting, coverage detects $70.0 \pm 5.3\%$ against $63.6 \pm 6.1\%$ for the deviation-seeking planner. The gap is stable across seeds.

Fig. 2 suggests why. The planner concentrates on the corridor, from which many elements are partially visible, and enters comparatively few rooms; coverage systematically enters every one. Two mechanisms plausibly contribute. First, with a uniform prior, maximising

expected information about every element’s deviation state is close to asking to see every element, which is what coverage does by construction — the objective may simply not be selective. Second, greedy benefit-per-metre is myopic: it repeatedly takes the locally cheapest informative viewpoint and pays for detours it cannot amortise, whereas a nearest-neighbour sweep is a serviceable travelling-salesman heuristic. The second mechanism points at the routing rather than the objective, and is the first thing we intend to fix.

C. The manipulator helps every planner

Freezing the arm costs the deviation-seeking planner 5.7 points ($63.6 \rightarrow 57.9$), coverage 6.4 ($70.0 \rightarrow 63.6$) and frontier 9.3 ($47.9 \rightarrow 38.6$); the goal-directed baseline is nearly unaffected because it barely stops (Fig. 4). In the selected plans the robot routinely parks once and adopts several postures, gathering additional observations for no traversal cost at all.

The honest reading is that the manipulator is valuable for as-built verification in general, not for our objective in particular. That is a weaker claim than we set out to make and a more useful one for practitioners choosing a platform.

VI. LIMITATIONS AND NEXT STEPS

The observation model is geometric; a depth-based detector behind the same interface is required before any claim about perception robustness. Scenes are synthetic, because the bounding-box IFC adapter would turn every doorway into a wall; a tessellating importer is needed for real models. Traversal cost reasons about walls, but the controller that executes a plan does not yet follow paths around them. Results are five seeds on one layout at one density; the density sweep the sampler was built for is not yet reported.

The immediate work is to replace greedy selection with tour construction and local improvement, which tests the routing explanation directly. If the planner still does not beat coverage, the uniform prior is implicated, and deviation-seeking planning will need a defensible reason to prefer some elements over others — from construction sequence, trade handover, or observed workmanship — rather than treating all modelled elements as equally suspect.

VII. CONCLUSION

We built the apparatus needed to ask whether deviation-seeking viewpoint planning pays, and used it to answer honestly: on this benchmark it does not yet beat systematic coverage once traversal distance is measured around walls, though including a manipulator in viewpoint selection helps every planner tested. We release the benchmark, the baselines and the evaluation so that the negative result can be attacked rather than repeated.

TABLE I

Deviations detected within a 60m budget (%), mean $\pm$ s.d. over five seeds). Ranking of the top two methods depends on how distance is measured.

Planner	Drivable	Straight-line
Coverage	70.0 ± 5.3	70.0 ± 5.3
Deviation-seeking (ours)	63.6 ± 6.1	72.1 ± 5.7
Frontier	47.9 ± 4.3	47.1 ± 6.1
Goal-directed	25.7 ± 4.2	42.9 ± 4.5

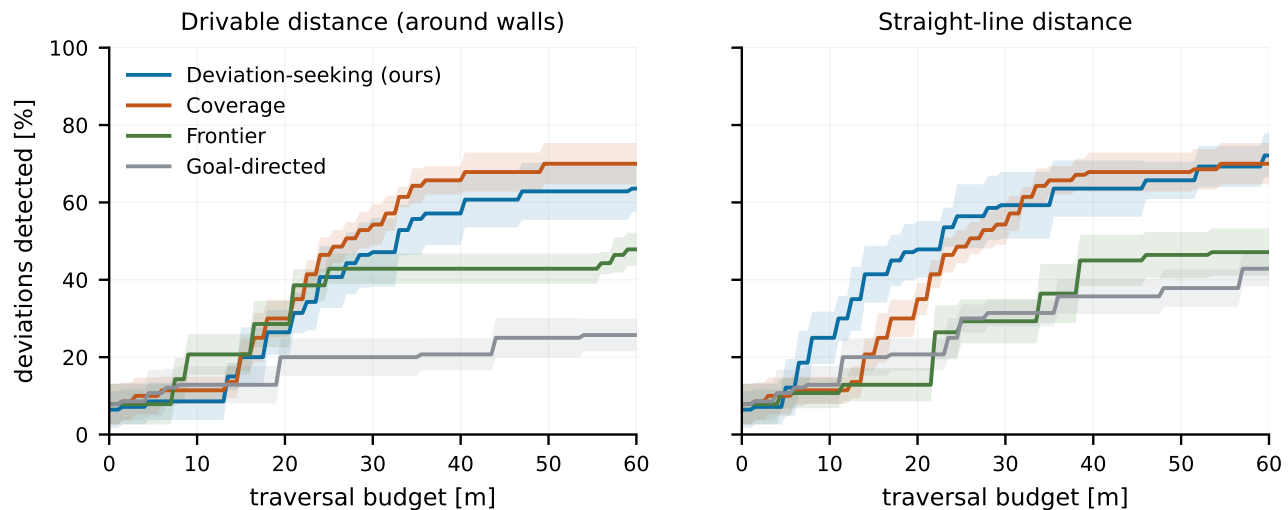


Fig. 3. Detections against traversal budget, mean over five deviation seeds with one standard deviation shaded. Left: drivable distance. Right: straight-line distance. The ranking of the top two methods reverses: measuring distance in straight lines credits the deviation-seeking planner with a lead it does not have.

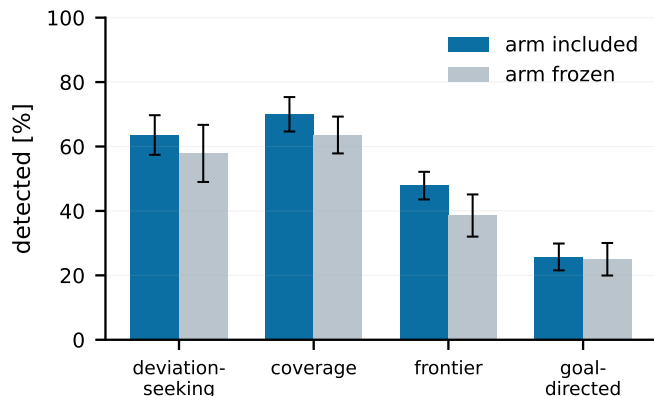


Fig. 4. Including the arm in viewpoint selection versus freezing it at one posture, under drivable distance. Every planner loses ground with a fixed sensor.

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